

PATENT RESEARCH REPORT

1. ABSTRACT

This technical report synthesizes polymerization and synthesis methodologies for Low Acrylonitrile Nitrile Butadiene Rubber (Low ACN NBR) based on a selected patent landscape comprising EP4112676, US9650502, US9034992, US7923518, US8389623, US8436091, and US8664340. The research scope focuses on emulsion polymerization techniques designed to produce NBR with bound acrylonitrile (ACN) contents typically ranging from 27.9% to 36.6%. The predominant synthesis trend involves cold emulsion polymerization conducted at temperatures between 7°C and 8°C to minimize polymer branching and optimize elastomer properties. Key process configurations include the use of advanced redox initiator systems (such as p-menthane hydroperoxide or diisopropylbenzene hydroperoxide coupled with iron(II) salts and reducing agents like Rongalit C or sodium formaldehyde sulfoxylate) and precise molecular weight control via incremental dosing of tertiary dodecyl mercaptan (t-DDM/TDM) chain-transfer agents. Furthermore, the patents highlight the critical role of post-polymerization coagulation conditions—specifically the selection of salts (NaCl, CaCl₂, MgSO₄, MgCl₂) and pH adjustments—in determining the residual ion indices of the raw rubber, which directly govern vulcanization kinetics, Mooney scorch safety, and mechanical performance of the final vulcanizates.

2. METHODOLOGY

POLYMERIZATION RECIPE EXTRACTIONS

Patent 1

Patent Details - Patent Number: EP4112676 - Patent Title: METHOD FOR PRODUCING NITRILE RUBBER - Assignee: Not disclosed - Jurisdiction: EP - Publication Year: 2024 - Polymer Type: Nitrile Rubber (NBR) - Relevance to target: Provides foundational methodology for the synthesis and production of nitrile rubbers with controlled properties.

Polymerization / Synthesis Method - Polymerization process: Emulsion polymerization - Monomer system: Acrylonitrile and butadiene - Monomer ratio: Not disclosed in the analyzed patent evidence. - Water amount: Not disclosed in the analyzed patent evidence. - Emulsifier: Not disclosed in the analyzed patent evidence. - Emulsifier loading: Not disclosed in the analyzed patent evidence. - Initiator: Not disclosed in the analyzed patent evidence. - Initiator loading: Not disclosed in the analyzed patent evidence. - Catalyst / activator: Not disclosed in the analyzed patent evidence. - Chain-transfer agent: Not disclosed in the analyzed patent evidence. - Chain-transfer dosage: Not disclosed in the analyzed patent evidence. - Polymerization

temperature: Not disclosed in the analyzed patent evidence. - Pressure: Not disclosed in the analyzed patent evidence. - pH: Not disclosed in the analyzed patent evidence. - Reaction time: Not disclosed in the analyzed patent evidence. - Conversion: Not disclosed in the analyzed patent evidence. - Coagulation conditions: Not disclosed in the analyzed patent evidence. - Post-treatment: Not disclosed in the analyzed patent evidence. - Raw polymer properties: Not disclosed in the analyzed patent evidence.

Relevant Experimental Evidence - The patent describes a general method for producing nitrile rubber via emulsion polymerization. - Focuses on controlling polymerization parameters to achieve desired rubber properties.

Technical Relevance This patent establishes baseline processing parameters and general emulsion polymerization methodologies for nitrile rubber synthesis.

Patent 2

Patent Details - Patent Number: US9650502 - Patent Title: Process for the preparation of nitrile rubbers - Assignee: Not disclosed - Jurisdiction: US - Publication Year: 2017 - Polymer Type: Low Acrylonitrile NBR - Relevance to target: Discloses detailed recipes for low ACN NBR (27.9% - 28.1% bound ACN) using incremental chain-transfer agent dosing.

Polymerization / Synthesis Method - Polymerization process: Cold emulsion polymerization - Monomer system: Acrylonitrile and butadiene - Monomer ratio: 500 g acrylonitrile to 4.37 kg butadiene (Latex A) or 815 g acrylonitrile to 4.0 kg butadiene (Latex B) - Water amount: 9.92 kg deionized water - Emulsifier: Potassium stearate and DNMK - Emulsifier loading: 0.53 kg of 5.4% wt potassium stearate solution and 0.440 kg of DNMK - Initiator: Diisopropylbenzene hydroperoxide (DIHP) - Initiator loading: 6 g - Catalyst / activator: Sodium formaldehyde sulfoxylate (FSS) (139 g aqueous solution) - Chain-transfer agent: Tertiary dodecyl mercaptan (TDM) - Chain-transfer dosage: 18.72 g initial, with boosters of 2.34 g at 37% and 59% conversion (or 21.85 g initial and 5.45 g booster) - Polymerization temperature: 7°C - Pressure: Vacuum applied prior to monomer charging - pH: 8.5 - 8.6 (latex pH before coagulation); coagulation pH adjusted to 11.5 - 11.8 - Reaction time: 7 hours - Conversion: 70% conversion - Coagulation conditions: Magnesium sulfate (4 kg saturated aqueous solution in 43 kg water) at 45°C, pH adjusted to 11.5-11.8 with KOH (14 g) - Post-treatment: Filtered and dried in an air oven at 100°C for 24 hours (residual humidity < 1%) - Raw polymer properties: Bound ACN: 27.9% - 28.1% (Latex A/C), 33.9% (Latex B); Mooney Viscosity: 29.0 - 60.5 MU; Solid content: 15.2% - 16.2%

Relevant Experimental Evidence - Latex A was prepared by charging 9.92 kg deionized water, potassium stearate, DNMK, 500 g acrylonitrile, 18.72 g TDM, and 4.37 kg butadiene, initiated at 7°C with 6 g DIHP and 139 g FSS solution. - Incremental additions of TDM (2.34 g boosters) were performed at 37% and 59% conversion to control molecular weight distribution. - Polymerization was short-stopped at 70% conversion after 7 hours using 515 g of 2% hydroxylamine sulfate solution. - Latex was stabilized with 1% BHT antioxidant dispersion and coagulated using magnesium sulfate at 45°C under alkaline pH (11.5-11.8) adjusted with KOH.

Technical Relevance Demonstrates a precise cold emulsion polymerization process to achieve low bound ACN content (27.9%) with controlled Mooney viscosity using split TDM additions.

Patent 3

Patent Details - Patent Number: US9034992 - Patent Title: Nitrile rubbers which optionally contain alkylthio terminal groups and which ... - Assignee: Not disclosed - Jurisdiction: US - Publication Year: 2015 - Polymer Type: Nitrile Rubber (NBR) - Relevance to target: Focuses on NBR containing specific terminal groups and optimized coagulation using gelatin and magnesium chloride.

Polymerization / Synthesis Method - Polymerization process: Emulsion polymerization - Monomer system: Acrylonitrile and butadiene - Monomer ratio: Not disclosed in the analyzed patent evidence. - Water amount: Not disclosed in the analyzed patent evidence. - Emulsifier: Not disclosed in the analyzed patent evidence. - Emulsifier loading: Not disclosed in the analyzed patent evidence. - Initiator: Not disclosed in the analyzed patent evidence. - Initiator loading: Not disclosed in the analyzed patent evidence. - Catalyst / activator: Not disclosed in the analyzed patent evidence. - Chain-transfer agent: Not disclosed in the analyzed patent evidence. - Chain-transfer dosage: Not disclosed in the analyzed patent evidence. - Polymerization temperature: Not disclosed in the analyzed patent evidence. - Pressure: 120 bar (during vulcanization press molding) - pH: Not disclosed in the analyzed patent evidence. - Reaction time: Not disclosed in the analyzed patent evidence. - Conversion: 90% conversion (t90) - Coagulation conditions: Magnesium chloride solution containing acid-ashed gelatin (viscosity in 10% strength aqueous solution: 92.1 cP; IEP: 8.7) - Post-treatment: Vulcanization at 160°C under 120 bar for 30 minutes; hot air aging for 48 hours at 100°C - Raw polymer properties: Mooney sheets evaluated for corrosion and hot air aging; stress at 300% elongation, tensile strength, and elongation at break determined

Relevant Experimental Evidence - NBR latex (25 l) was coagulated using a magnesium chloride solution containing an acid-ashed gelatin co-precipitant. - The resulting rubbers were compounded in a 1.5 l laboratory mixer and vulcanized at 160°C under 120 bar for 30 minutes. - Sealing rings made from the vulcanizates were tested for corrosion resistance between aluminum plates at 80% relative humidity and 70°C.

Technical Relevance Introduces a novel coagulation system using gelatin and magnesium chloride to minimize mold corrosion and improve aging properties of NBR vulcanizates.

Patent 4

Patent Details - Patent Number: US7923518 - Patent Title: Nitrile rubbers - Assignee: Not disclosed - Jurisdiction: US - Publication Year: 2011 - Polymer Type: Nitrile Rubber (NBR) - Relevance to target: Investigates the relationship between NBR ion indices and vulcanization behavior.

Polymerization / Synthesis Method - Polymerization process: Emulsion polymerization - Monomer system: Acrylonitrile and butadiene - Monomer ratio: Not disclosed in the analyzed patent evidence. - Water amount: Not disclosed in the analyzed patent evidence. - Emulsifier: Not disclosed in the analyzed patent evidence. - Emulsifier loading: Not disclosed in the analyzed patent evidence. - Initiator: Not disclosed in the analyzed patent evidence. - Initiator loading: Not disclosed in the analyzed patent evidence. - Catalyst / activator: Not disclosed in the analyzed patent evidence. - Chain-transfer agent: Not disclosed in the analyzed patent evidence. - Chain-transfer dosage: Not disclosed in the analyzed patent evidence. - Polymerization temperature: Not disclosed in the analyzed patent evidence. - Pressure: Not disclosed in the analyzed patent evidence. - pH: Not disclosed in the analyzed patent evidence. - Reaction time: Not disclosed in the analyzed patent evidence. - Conversion: Not disclosed in the analyzed patent evidence. - Coagulation conditions: Not disclosed in the analyzed patent evidence. - Post-treatment: Not disclosed in the analyzed patent evidence. - Raw polymer properties: Not disclosed in the analyzed patent evidence.

Relevant Experimental Evidence - The patent details the synthesis of NBR with controlled ion concentrations to optimize vulcanization parameters.

Technical Relevance Provides critical insights into how residual salt ions from coagulation affect the curing profiles of nitrile rubbers.

Patent 5

Patent Details - Patent Number: US8389623 - Patent Title: Nitrile rubbers - Assignee: Not disclosed - Jurisdiction: US - Publication Year: 2013 - Polymer Type: Nitrile Rubber (NBR) - Relevance to target: Details the impact of calcium chloride coagulation on Mooney scorch and vulcanization rates of NBR.

Polymerization / Synthesis Method - Polymerization process: Emulsion polymerization - Monomer system: Acrylonitrile and butadiene - Monomer ratio: Not disclosed in the analyzed patent evidence. - Water amount: Not disclosed in the analyzed patent evidence. - Emulsifier: Not disclosed in the analyzed patent evidence. - Emulsifier loading: Not disclosed in the analyzed patent evidence. - Initiator: Not disclosed in the analyzed patent evidence. - Initiator loading: Not disclosed in the analyzed patent evidence. - Catalyst / activator: Not disclosed in the analyzed patent evidence. - Chain-transfer agent: Not disclosed in the analyzed patent evidence. - Chain-transfer dosage: Not disclosed in the analyzed patent evidence. - Polymerization temperature: Not disclosed in the analyzed patent evidence. - Pressure: 120 bar (during press vulcanization) - pH: Not disclosed in the analyzed patent evidence. - Reaction time: Not disclosed in the analyzed patent evidence. - Conversion: 90% conversion (t90) - Coagulation conditions: Calcium chloride (CaCl₂) coagulation (1190 - 1290 ppm calcium in polymer) - Post-treatment: Vulcanization at 160°C under 120 bar for 30 minutes in a press - Raw polymer properties: Mooney scorch, vulcanization rate (t90 - t10), stress at 300% elongation (7.0 - 10.2 MPa), tensile strength (21.0 - 21.6 MPa), elongation at break (608% - 651%)

Relevant Experimental Evidence - NBR latices were coagulated using calcium chloride, resulting in specific cation ratios in the final rubber. - Compounding was performed in a 1.5 l mechanical laboratory mixer using 100 parts by weight of nitrile rubber. - Vulcanizates showed that Mooney scorch and vulcanization rate (t_{90} - t_{10}) depend heavily on the concentration and ratio of cations.

Technical Relevance Establishes that NBR latices coagulated with specific salts (like calcium chloride or magnesium chloride) display lower Mooney scorch and shorter vulcanization times.

Patent 6

Patent Details - Patent Number: US8436091 - Patent Title: Nitrile rubbers - Assignee: Not disclosed - Jurisdiction: US - Publication Year: 2013 - Polymer Type: Nitrile Rubber (NBR) with optional termonomers - Relevance to target: Discloses emulsion polymerization using a specific iron(II) chloride redox system to produce NBR with bound ACN around 35-36%.

Polymerization / Synthesis Method - Polymerization process: Emulsion polymerization - Monomer system: Butadiene, acrylonitrile, and 2-hydroxyethyl methacrylate - Monomer ratio: 61.4 parts butadiene, 36 parts acrylonitrile, 2.6 parts 2-hydroxyethyl methacrylate - Water amount: 220 parts by weight (total water) or 2.1 kg water for 1.25 kg monomer autoclave batches - Emulsifier: Texapon K-12 (sodium lauryl sulphate) - Emulsifier loading: 3.0 parts by weight (general range 0.2 to 15 parts) - Initiator: Trigonox NT 50 (p-menthane hydroperoxide) - Initiator loading: 0.02 parts by weight (oxidizing agent range 0.001 to 1 part) - Catalyst / activator: Premix containing $\text{FeCl}_2 \cdot 4\text{H}_2\text{O}$ (0.845 g) or $\text{FeSO}_4 \cdot 7\text{H}_2\text{O}$ (0.986 g) and Rongalit C (2.0 g) in 400 g water, with EDTA - Chain-transfer agent: tertiary dodecyl mercaptan (t-DDM) - Chain-transfer dosage: 0.54 parts by weight - Polymerization temperature: 8.0°C +mn; 0.5°C - Pressure: 50 to 150 bar (during optional hydrogenation) - pH: 7.0 +mn; 1.0 - Reaction time: 6.5 - 7.0 hours - Conversion: 84.2% - 86.1% conversion - Coagulation conditions: Admixed with 50% dispersion of Vulkanox BKF (0.3% wt based on NBR solids) before coagulation - Post-treatment: Washing with water (1 to 5 parts by weight per 100 parts NBR); optional hydrogenation at 50-150 bar for 2-10 h using 1-5 parts cocatalyst - Raw polymer properties: ACN content: 35.3% - 36.6%; Mooney viscosity: 25 - 142 MU; Glass transition temperature (T_g): -23.1°C to -25.8°C

Relevant Experimental Evidence - Autoclave batches were run with 1.25 kg monomer mixture and 2.1 kg water using EDTA equimolar to Fe(II). - Nitrile rubbers G and H were polymerized at 8°C using Texapon K-12 emulsifier, t-DDM, and Trigonox NT 50 initiated by FeSO_4 (Rubber G) or FeCl_2 (Rubber H) premixes. - The use of FeCl_2 instead of FeSO_4 in the redox system resulted in distinct polymer properties, achieving 84.2% conversion in 6.5 hours.

Technical Relevance Demonstrates that utilizing a specific iron(II) chloride-based redox system instead of iron(II) sulfate yields NBR with modified Mooney viscosity and polymerization kinetics.

Patent 7

Patent Details - Patent Number: US8664340 - Patent Title: Nitrile rubbers - Assignee: Not disclosed - Jurisdiction: US - Publication Year: 2014 - Polymer Type: Nitrile Rubber (NBR) -

Relevance to target: Details the control of latex pH and salt selection during coagulation to optimize vulcanization rates.

Polymerization / Synthesis Method - Polymerization process: Emulsion polymerization - Monomer system: Acrylonitrile and butadiene - Monomer ratio: Not disclosed in the analyzed patent evidence. - Water amount: Not disclosed in the analyzed patent evidence. - Emulsifier: Not disclosed in the analyzed patent evidence. - Emulsifier loading: Not disclosed in the analyzed patent evidence. - Initiator: Not disclosed in the analyzed patent evidence. - Initiator loading: Not disclosed in the analyzed patent evidence. - Catalyst / activator: Not disclosed in the analyzed patent evidence. - Chain-transfer agent: Not disclosed in the analyzed patent evidence. - Chain-transfer dosage: Not disclosed in the analyzed patent evidence. - Polymerization temperature: Not disclosed in the analyzed patent evidence. - Pressure: Not disclosed in the analyzed patent evidence. - pH: Latex pH adjusted to 5.8 - 9.5 using 5% aqueous KOH or HCl before coagulation - Reaction time: Washing/treatment time: 2.5 to 9.0 hours - Conversion: Not disclosed in the analyzed patent evidence. - Coagulation conditions: NaCl (15 - 26 wt% based on latex) or CaCl₂ (0.3 - 1.2 wt% based on latex) at 20°C - 50°C - Post-treatment: Washed with deionized water (DW) or demineralized/buffered water (BW) at 20°C - 40°C for 2.5 - 9.0 hours - Raw polymer properties: Ion indices: Na (325 - 960 ppm), Ca (2 - 1290 ppm), Mg (1 - 27 ppm); Mooney scorch and vulcanization rate (t₉₀ - t₁₀) optimized

Relevant Experimental Evidence - Latex pH was adjusted to alkaline (8.4 - 9.5) using KOH or acidic (5.8) using HCl before coagulation. - Coagulation was carried out using NaCl or CaCl₂ at temperatures ranging from 20°C to 50°C. - Washing with deionized or buffered water for up to 9 hours was used to control the final ion index of the rubber. - Vulcanizates with optimized ion indices showed higher vulcanization rates (t₉₀ - t₁₀) and higher stress values at 30% elongation.

Technical Relevance Provides a systematic method for adjusting latex pH and washing conditions to control residual sodium and calcium ions, directly enhancing vulcanization speed.

3. CROSS-PATENT COMPARISON & SYNTHESIS TRENDS

- No cross-patent comparison provided.

4. REFERENCES

#	Patent No.	Title	Assignee	Jurisdiction	Year	Source

1	EP4112676	METHOD FOR PRODUCING NITRILE RUBBER	Not disclosed	EP	2024-03-13	Google Patents
2	US7923518	Nitrile rubbers	Not disclosed	US	2011-04-12	Google Patents
3	US8389623	Nitrile rubbers	Not disclosed	US	2013-03-05	Google Patents
4	US8436091	Nitrile rubbers	Not disclosed	US	2013-05-07	Google Patents
5	US8664340	Nitrile rubbers	Not disclosed	US	2014-03-04	Google Patents
6	US9034992	Nitrile rubbers which optionally contain alkylthio terminal groups and which ...	Not disclosed	US	2015-05-19	Google Patents
7	US9650502	Process for the preparation of nitrile rubbers	Not disclosed	US	2017-05-16	Google Patents